De La Salle University - Dasmariñas
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	**:**	**S-ITNT311LA**
COURSE TITLE	**:**	**Internet of Things**
COURSE TYPE	**:**	Laboratory
COURSE CREDIT	**:**	1 Unit
LEARNING MODALITY	**:**	Fully Onsite
PRE-REQUISITES	**:**	
CO-REQUISITES	**:**	**S-ITNT311**
PROFESSOR	:	Rochelle D. Sermaña Monday 1000-1200 Online Tuesday 1300-1600 COS100-C Wednesday 1600-1700 Online Thursday 1500-1600 Online rdsermana@dlsud.edu.ph (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

This course, the Internet of Things, IoT, describes a growing industry of digital technology being harnessed to the Internet in ways that will improve the lives of every person on this planet. It provides an overview of key concepts and challenges related to digital transformation. The course examines the evolution of the Internet and how the interconnection of people, processes, data and things is transforming every industry.

The introduction to the Internet of Things course explains what the IoT is, what it does, how it is part of digital transformation, and how students can become part of this. Moreover, students will also learn about the exponential increase of intelligent devices connected to the Internet and will learn to program one of these intelligent devices. The course explains artificial intelligence and the impact of automation to our future. Lastly, students will understand the increased importance of privacy and security.

COURSE LEARNING OUTCOMES:

By the end of this course, students are expected to:
CLO1. Explain the meaning and impact of digital transformation.
CLO2. Apply basic programming to support IoT devices.
CLO3. Examine how data provides value to digital business and society.
CLO4. Explore the benefits of automation in the digitized world.
CLO5. Distinguish the need for security in the digitized world.
CLO6. Evaluate opportunities provided by digital transformation.
CLO7. Demonstrate 21st century skills in all learning activities.

LEARNING PLAN

Learning Outcomes for Midterm Period

Course Learning Outcomes	Topic Learning Outcomes
CLO1. Explain the meaning and impact of digital transformation.	• TLO1. Understand the concept of digital transformation. • TLO2. Identify devices that connect to the IoT
CLO2. Apply basic programming to support IoT devices.	• TLO3. Apply basic programming to support IoT devices. • TLO4. Transforming ideas into prototypes
CLO3. Examine how data provides value to digital business and society.	• TLO5. Understanding everything about big data.
CLO7. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	**Date**	**Teaching-Learning Activities/Assessment Strategy**	**Technology Enabler**	**Allotted Hours**
Module 0	Week 1 (Aug. 19-Aug. 24	• Gospel Reading, Reflection & Prayer • Presentation of Module 0**:** Course Introduction • Discussion of Course requirement details and Laboratory Policies	Schoolbook MS Teams Canva	3

Module 1 TLO 1 TLO 2 CLO 1 CLO 7	Week 2 (Aug. 26 - Aug. 31)	• Gospel Reading, Reflection & Prayer • Discussion of Module 1: Everything is Connected • Activity: How Connected Are You?	Schoolbook MS Teams Canva	3
	Week 3 (Sept. 2 - Sept. 7)	• Gospel Reading, Reflection & Prayer • Continuation of Module 1 • Activity: Packet Tracer: Deploying and Cabling Devices • Review Module 1 Terms and Concepts		3
Module 2 TLO 3 TLO 4 CLO 2 CLO 7	Week 4 (Sept. 9 - Sept. 14)	• Gospel Reading, Reflection & Prayer • Discussion of Module 2: Everything becomes Programmable. • Activity: Create a Process Flowchart • Enabling assessment: Blinking an LED Using Blockly	Schoolbook MS Teams Canva	3
	Week 5 (Sept. 16- Sept. 21)	• Gospel Reading, Reflection & Prayer • Continuation discussion of programming language concepts • Activity: Create a Simple Game Using Python • Enabling assessment: Quiz on Module 2		3
Module 3 TLO 5 TLO 6 CLO 3 CLO 4 CLO 7	Week 6 (Sept. 23 - Sept. 28)	• Gospel Reading, Reflection & Prayer • Discussion of Module 3: Everything Generates Data • Activity: Explore a Large Dataset • Enabling assessment: Quiz on Module 3 •	Schoolbook MS Teams Canva	3
	Week 7 (Sept. 30 - Oct. 5)	• Gospel Reading, Reflection & Prayer • Activity: Using Excel to Forecast • Self-paced reading (Review of Concepts from Modules 1-3)		3
Module 1 to 3	Week 8 (Oct. 7 - Oct. 12)	• Summative assessment		3
			TOTAL	**24**

Learning Outcomes for Final Period

Course Learning Outcomes	Topic Learning Outcomes
CLO4. Explore the benefits of automation in the digitized world.	• TLO6. Identify everything that can be automated.
CLO5. Distinguish the need for security in the digitized world.	• TLO7. Recognize and understand the importance of security in the digitized world
CLO6. Evaluate opportunities provided by digital transformation.	• TLO8. Recognize educational and business opportunities.
CLO7. Demonstrate 21st century skills in all learning activities.	• *Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real-World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcome(s)	Date	Teaching-Learning Activities/Assessment Strategy	Technology Enabler	Allotted Hours
Module 4 TLO 5 TLO 6 CLO 3 CLO 4 CLO 7	Week 10 (Oct. 21 - Oct. 26)	• Gospel Reading, Reflection & Prayer • Discussion of Module 4: Everything Can be Automated. • Activity: Packet Tracer: Explore the Smart Home	Schoolbook MS Teams Canva	3
	Week 11 (Oct 28 - Nov. 2)	• Gospel Reading, Reflection & Prayer • *Independent Learning Week*		3
Module 5 TLO 7 TLO 8 CLO 5 CLO 6 CLO 7	Week 12 (Nov. 4 - Nov. 9)	• Gospel Reading, Reflection & Prayer • Discussion of Module 5: Everything needs to be Secured. • Activity: Packet Tracer: Configure Wireless Security	Schoolbook MS Teams Canva	3
	Week 13 (Nov. 11 - Nov. 16)	• Gospel Reading, Reflection & Prayer • Final output comprehensive instruction • Enabling assessment: Quiz on Module 5		3
Module 6 TLO 6 TLO 7 TLO 8 TLO 9 CLO 3 CLO 4 CLO 5 CLO 6	Week 14 (Nov. 18 - 23)	• Gospel Reading, Reflection & Prayer • Activity: IoT Related Job and Opportunities	Schoolbook MS Teams Canva	3
	Week 15 (Nov. 25 - Nov. 30)	• Gospel Reading, Reflection & Prayer • Preparation time/ Consultation (if able and willing)		3
	Week 16 *Dec. 2 - Dec. 7)	• Gospel Reading, Reflection & Prayer • Final Output Presentation (Summative Assessment 2)		3
	Week 17 (Dec. 9 - Dec. 14)	• Summative assessment • Reflection of the entire module • Submission of Final Output		3
			TOTAL:	**24**

GRADING SYSTEM

Midterm

Enabling Assessments - 40%
Summative Assessments - 60%
100%

Finals

Enabling Assessments - 40%
Summative Assessments - 60%

100%

Overall Final Grade = Midterm + Final

COURSE POLICIES AND REQUIREMENTS

1. This course follows the Fully Onsite Modality. All activities will be done in the laboratory classroom.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar.

3. Laboratory Policies
   a. Students must always be honest; cheating and plagiarism in any form will merit a grade of 0.00.
   b. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless a special permission is sought. Cellular phones also cannot be used as calculator during examination.
   d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

4. Communication and Feedback
   a. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
      i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
      ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through Schoolbook messaging feature.
      iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
   b. No use of foul, curse, and other inappropriate words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

5. Attendance Policy
   a. Attendance Policy for Onsite Classes is scheduled based on the learning plan.
   b. The onsite meeting will only be held at the schedule indicated in the registration form.
   c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

6. Intellectual Property Policy
   a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
   b. Using any illegally obtained software is prohibited.
   c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
   d. Giving of passwords to Schoolbook and Microsoft 365 is prohibited. Likewise, accessing Schoolbook and Microsoft 365 accounts other than the students' personal account is also

strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

7. Other Policies
    a. Attire - students must wear the prescribed uniform, decent casual or business attire prescribed by the Student Handbook.

REFERENCES

Onsite References

Provider	Reference Material
AEA Library TK 5105.8857 .L615 2019	Liberg, Olof. 2019. Cellular Internet of things : technologies, standards, and performance. (San Diego, CA, United States): Academic Press
AEA Library LB 2395.7 .Sl36 2019	Mickey Slimp. 2019. How the Internet of things is changing our colleges, our classrooms, and our students. (Lanham, Maryland) : Rowman & Littlefield

Online References

Call number or e-provider	Reference Material
AEA EBSCO Portal Accession Number: 2013945	Gacovski, Zoran. Ashland : Arcler Press. 2019. Internet of Things. eBook., Database: eBook Academic Collection (EBSCOhost)
AEA EBSCO Portal Accession Number: 503597	Jan Holler; Vlasios Tsiatsis; Catherine Mulligan; Stamatis Karnouskos; Stefan Avesand; David Boyle. Burlington : Academic Press. 2014. Internet of Things. eBook., Database: eBook Academic Collection (EBSCOhost)
AEA EBSCO Portal Accession Number: 1775071	Peter Waher. Birmingham : Packt Publishing. 2018. Mastering Internet of Things : Design and Create Your Own IoT Applications Using Raspberry Pi 3. eBook., Database: eBook Academic Collection (EBSCOhost)
AEA EBSCO Portal Accession Number: 1874792	Shishir K. Shandilya; Soon Ae Chun; Smita Shandilya. Series: River Publishers Series in Information Science and Technology. Aalborg : River Publishers. 2018. Internet of Things Security : Fundamentals, Techniques and Applications. eBook., Database: eBook Academic Collection (EBSCOhost)
AEA EBSCO Portal Accession Number: 1226259	Watts, Silvia. Series: Internet Theory, Technology and Applications. New York : Nova Science Publishers, Inc. 2016. The Internet of Things (IoT) : Applications, Technology, and Privacy Issues. eBook., Database: eBook Academic Collection (EBSCOhost)
Website	Introduction to IoT. 2018 Jan 22. Networking Academy. [Accessed: 2024 August 6]. https://www.netacad.com/courses/iot/introduction-iot.
Website	Velazquez R. 2022 Sep 22. What Is The Internet of Things? A Complete Guide To IoT. | Built In. builtincom. [Accessed: 2024 August 6]. https://builtin.com/internet-things.

Prepared: **Information Technology Department**
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Approved: **(SGD) Ms. Sherry B. Naz**
Chair, ITD

Endorsed: **(SGD) Ms. Marivic R. Mitschek**
Dean, CICS